Shinjan Goswami

Bachelor of Statistical Data Science
Indian Statistical Institute, Kolkata

Phone: +91 7047486116
shinjangoswami@gmail.com
LinkedIn: linkedin.com/in/shinjangoswami
Portfolio Website: shinjangoswami.github.io/

ABOUT ME

I am currently pursuing a Bachelors in Statistical Data Science at Indian Statistical Institute Kolkata. I am more comfortable working with various mathematical and probabilistic models, and implementing them in the scope of Computer Science. I am comfortable with Python, Java and R languages and have learnt them over my 10+2 level schooling years. I am looking forward to contributing in Computer Science, Mathematics and Machine Learning fields.

EDUCATION

Indian Statistical Institute, Kolkata — 2025–2029 (exp.)
Bachelor of Statistical Data Science — 77.5%

Burdwan Model School, Purba Bardhhaman — 2025
Class XII (Senior Secondary), CBSE — 92.2%

St.Xavier's School, Burdwan — 2023
Class X (Secondary), ICSE — 98.2%

SCHOLASTIC ACHIEVEMENTS

- **Rank 50** in **ISI BSDS Merit List** [2025]
- **Rank 1975** in **IISER Aptitude Test** [2025]
- **Rank 954** in **WBJEE** [2025]
- **Certificate of Excellence from Burdwan Model School** for **result in AISSCE examinations** [2025]
- **Junior Talent Search Encouragement Awardee** in **Jagadis Bose National Science Talent Search (JBNSTS)** [2023]

SKILLS & INTERESTS

- **Programming Languages**: Python, R, Java
- **Tools**:GitHub, SQL, RStudio, OverLeaf
- **Concepts**: Optimization, Numerical Methods,Statistical Inference, Simple Linear and Multivariate Regression Techniques, Probability Theory, Data Structures and Algorithms
- **Interests**: Artificial Intelligence, Mathematics, Deep Learning, Quantitative Finance, Computer Science Engineering
- **Spoken Languages**: English, Hindi, Bengali

EXTRA CURRICULAR EDUCATIONAL COURSES

- **Computer Networks and Internet Protocol** | **NPTEL** [Jun 2026]
  - Completed the course comprising of assignments and a final offline exam at a designated NPTEL centre. Scored a combined grade of 67% and NPTEL Elite Cerificate.
- **Computer Architecture and Organization** | **NPTEL** [Nov 2026]
  - Completed the course comprising of assignments and a final offline exam at a designated NPTEL centre. Scored a combined grade of 73% and NPTEL Elite Cerificate.

PROJECTS

- **Facial Recognition Using Eigenfaces and Fisherfaces**
  - Implemented and compared two classical face-recognition methods using Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) for dimensionality reduction and discriminative feature extraction.
  - Developed an image-preprocessing and recognition pipeline involving grayscale conversion, resizing, histogram equalization, pixel standardization, and cosine-similarity ranking.
  - Evaluated the methods on 38 identities using 304 training images and 76 test images; achieved 94.74% Rank-1 accuracy with Eigenfaces and 97.37% with Fisherfaces. Fisherfaces also achieved 100% Top-5 accuracy.

- **Project Instructor:** Prof. Arnab Chakraborty
- **Collaborators:** Shaurya Swaraj, Swapnil Mondal
- **Repository:** GitHub

- **Count-Min Sketch: Frequency Estimation in Large Data Streams**
  - Studied Count-Min Sketch, a probabilistic data structure for approximate frequency estimation in large-scale data streams.
  - Analysed the role of hash functions and collisions in frequency estimation, deriving probabilistic error bounds using expectation and Markov's inequality.
  - Examined the trade-off between estimation accuracy and memory usage, along with applications in network traffic analysis, heavy-hitter detection, and web analytics.
  - **Project Instructor:** Prof. Arnab Chakraborty
  - **Collaborators:** Shaurya Swaraj, Swapnil Mondal, Tridham Sai Dudham, Akshay Yenni, Daksh Kumar

RELEVANT COURSES

- **Statistics & Probability:** Probability Theory, Statistical Inference, Regression & Linear Statistical Models
- **Data Science & Computing:** Data Structures & Algorithms, Data Analysis using R and Python
- **Mathematics:** Linear Algebra, Multivariate Calculus, Differential Equations, Real Analysis, Optimization & Numerical Methods, Graph Theory
- **Economics:** Microeconomics, Macroeconomics